

EDUCATION

- National Institute of Technology Durgapur

Bachelor of - Computer Science and Engineering, CGPA: 8.01

Durgapur, West Bengal, India

Sep 2024- Present

Semester	SGPA	Aggregate CGPA
Semester 1	8.35	8.35
Semester 2	7.35	7.88
Semester 3	8.15	7.99
Semester 4	8.09	8.01
- Jawahar Navodaya Vidyalaya

(10) CBSE Boards; Percentage: 97.6

Sarang, Dhenkanal, Odisha, India

August 2020 – April 2022
- Jawahar Navodaya Vidyalaya

(12) CBSE Boards; Percentage: 92

Ozhar, Barwani, Madhya Pradesh, India

April 2022 – April 2024

PROFILE SUMMARY

Description:

Driven data enthusiast with a strong foundation in **Data Science, Machine Learning, and Deep Learning**, complemented by hands-on experience in **feature engineering, data extraction, and data preprocessing**. Proficient in **C, C++, Python, JS and SQL**, with a solid grasp of building and optimizing data driven solutions. Experienced in developing projects across domains, including cloud-based systems and intelligent applications, with working knowledge of full-stack development to support end-to-end solution building.

Technical Skills:

- * **Languages:** C, C++, Python, JavaScript
- * **Web Development:** React.js, Node.js, Express.js, FastAPI.
- * **Data Science & AI:** Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision (Open CV), Time Series Analysis, EDA
- * **Libraries & Frameworks:** Tensorflow, Keras, Scikit-learn, Pandas, NumPy.
- * **Databases, Cloud & DevOps:** PostgreSQL, MySQL, AWS, Docker.
- * **Core Subjects:** Object-Oriented Programming (OOPs),DBMS, OS.

Non-Technical Skills:

Leadership, Communication skills, Problem solving skills, Team management, Adaptability & Learning Agility.

EXPERIENCE

- Senior web Developer at Radio NITroz - Feb 2025- Present: [\[github\]](#) [\[Deployment\]](#)

Description:

Developing Websites for The Official Entertainment Club Of NIT Durgapur
- Senior DataScience and AIML model Developer at R&D Cell NIT Durgapur - Feb 2025- Present:

Description:

Developing Full stack integrated Machine Learning projects, Conducting Workshops For Freshers Teaching them Fundamentals Of ML.

PROJECTS

- FinAI - Stock Analysing Chatbot [\[github\]](#) [\[Deployment\]](#):

o Description:

- * **Architected a stateful Multi-Agent System (MAS)** using LangGraph and Groq (Llama-3.3-70b) to build a production-grade, ultra-low latency investment research platform.
 - * **Engineered an intent-driven orchestrator** that dynamically routes queries across 5 specialized execution pipelines (Vision, Fundamental, Portfolio, etc.) to optimize compute and response times.
 - * **Designed strict anti-hallucination guardrails**, integrating an adversarial AI debate panel to penalize sycophancy, and a deterministic Pandas and NumPy sandbox to compute portfolio risk metrics without relying on LLM arithmetic.

o Tech Stack:

Python, LangChain, LangGraph, Streamlit

- **StarRoute – Distributed Space Travel Platform** [\[github\]](#):
 - *Description:*
 - * **Developed a full-stack distributed booking platform** scaling a C++ OOP model into a highly concurrent web application to solve the **Thundering Herd** problem during high-traffic ticket sales.
 - * **Engineered a high-performance C++ REST API** using the Drogon framework and PostgreSQL, applying OOP principles (Polymorphism, Singleton) for role-based passenger management and encapsulated ticket pricing.
 - * **Architected an asynchronous booking engine** utilizing Redis distributed locks (SETNX) to prevent seat double-booking and RabbitMQ message queues for non-blocking background transaction processing.
 - * **Built a responsive React SPA** with Vite, featuring custom interactive space-themed seat maps, dynamic state hydration, and asynchronous polling for real-time boarding pass generation.
 - **Tech Stack:** C++, Drogon, PostgreSQL, Redis, RabbitMQ, React.js, Vite
- **LoclyAI – Smart Civic Issue Reporting Platform** [\[github\]](#) [\[Deployment\]](#):
 - *Description:*
 - * **Engineered an automated computer vision pipeline** utilizing a YOLOv8 model to accurately detect, localize, and categorize user-reported civic anomalies (e.g., potholes, garbage accumulation, broken traffic signals).
 - * **Architected a decoupled microservices inference system**, using Node.js/Express.js as an API gateway to manage client requests and automated geo-tagging, seamlessly routing data to a high-performance **FastAPI (Python)** server for low-latency ML predictions.
 - * **Developed a responsive frontend interface** using React.js to streamline the civic reporting workflow, allowing users to upload image evidence, visualize mapped anomalies, and track resolution statuses in real-time.
 - **Tech Stack:** Python, FastAPI, YOLOv8 (Computer Vision), React.js, Node.js, Express.js, PostgreSQL, REST APIs

POSITION OF RESPONSIBILITY & EXTRA-CURRICULARS

Position of Responsibility

- **Core Organising Committee member** [@AarohanNITDGP](#).
- **Senior Web Developer** [@RadioNITrozNITDGP](#).
- **Artificial intelligence and Machine Learning Developer in Research and Development cell** [@CCANITDGP](#).

Extra-Curriculars

- Blogging.
- Playing Cricket etc.
- Watching movies.
- Trading and analysing Stocks .

Achievements

- Secured an All India Rank of **10,281** in the **JEE Mains** entrance examination.
- Selected for the on-campus round of **Smart India Hackathon (SIH) 2024**.
- Awarded **Best Speaker** in the Youth Parliament at NIT Durgapur.